CSE110: Programming Language I (Lab)
Fall 2025
Lab Exam - 04
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

A teacher records the attendance of students using their roll numbers, but due to manual entry, the roll numbers are not arranged in a proper order. Write a Java program that should accept the **number of students present**, store their roll numbers in an **integer array** and display the **original attendance list**. After that, the program must **rearrange the roll numbers** so that **even numbers appear first followed by the odd numbers** by maintaining the **original order**. Finally, the program should **display** the **reordered attendance list**, along with the **counts** for **even roll numbers and odd numbers**, the **highest** and **lowest roll numbers.**

Constraints:

- The reordered list must be out-of-place.
- You cannot use any built-in function other than .length.

Sample Input	Sample Output
Enter number of students present: 8 Enter roll numbers: Student 1: 15 Student 2: 8 Student 3: 21 Student 4: 4 Student 5: 10 Student 6: 7 Student 7: 18 Student 8: 3	Original attendance list: 15 8 21 4 10 7 18 3 Reordered attendance list: 8 4 10 18 15 21 7 3 Number of even roll numbers: 4 Number of odd roll numbers: 4 Highest roll number: 21 Lowest roll number: 3
Explanation : In the sample input, the teacher enters **8 students** and their roll numbers: 15, 8, 21, 4, 10, 7, 18, and 3. The program first displays these as the original attendance list. It then rearranges the roll numbers by placing all **even numbers first** (8, 4, 10, 18) followed by all **odd numbers** (15, 21, 7, 3), while keeping their original order. The program counts **4 even** and **4 odd** roll numbers. Finally, it identifies **21** as the highest roll number and **3** as the lowest roll number.	